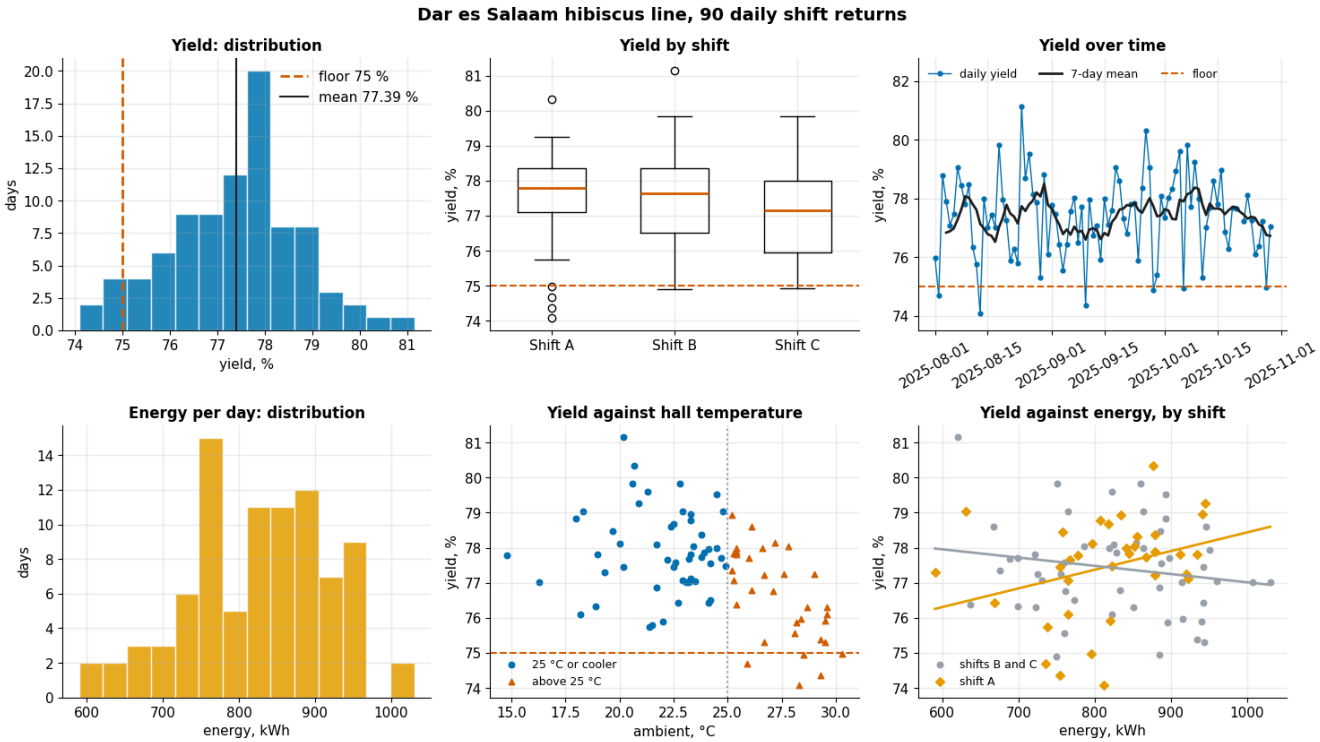
A03 worked exemplar: process capability notebook for Dar es Salaam hibiscus

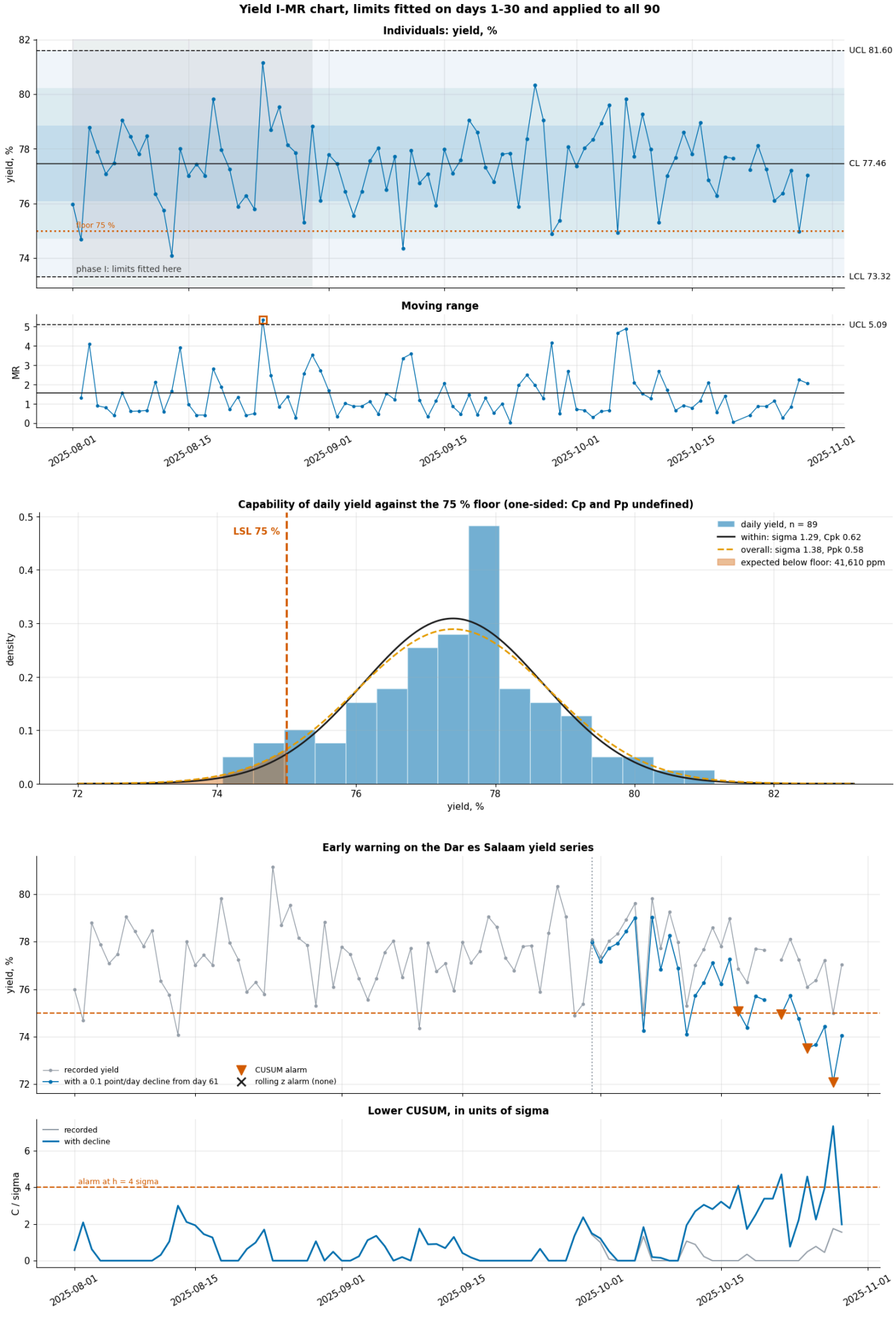
This is a worked example of an A03 submission, made by the programme team to show a notebook at the upper bands. It is set at Ubuntu Naturals' Dar es Salaam primary-processing plant, where hibiscus calyx is dried, ground and sieved for the Hibiscus Pressure tablets, and not at the Port Louis liquid-fill line your brief describes. Dar es Salaam keeps one hand-typed yield figure a day against a one-sided floor. Port Louis gives you a temperature reading every second and fill volumes in subgroups of five against limits on both sides, so the charts, the capability indices and the early-warning detector you need are different ones. Do not copy it. The marks above Pass go to analysis fitted to your own data and to Spirulina Power, and code or wording lifted from a daily yield record answers questions your datasets do not ask.

The submission

The submission is `A03_exemplar_dar_es_salaam_hibiscus_process_capability.ipynb`, an executed notebook with its outputs saved. A learner's file would be named `A03_Process_Capability_<Surname>.ipynb`; the exemplar keeps the programme's exemplar naming instead. The notebook reads the shift returns by a path relative to its own folder, so it runs from where it is saved. `A03_exemplar_dar_es_salaam_hibiscus_process_capability.html` is the same notebook rendered for reading without Jupyter, and `A03_exemplar_dar_es_salaam_hibiscus_figures.json` holds every figure the narrative states, as the notebook's last cell printed them.

The previews below are four of the notebook's own figures, taken from its executed outputs.




Yield I-MR chart, limits fitted on days 1-30 and applied to all 90
Individuals: yield, %
UCL 81.60
CL 77.46
LCL 73.32
floor 75 %
phase I: limits fitted here
yield, %
Moving range
UCL 5.09
MR
2025-08-01
2025-08-15
2025-09-01
2025-09-15
2025-10-01
2025-10-15
2025-11-01
Capability of daily yield against the 75 % floor (one-sided: Cp and Pp undefined)
LSL 75 %
daily yield, n = 89
within: sigma 1.29, Cpk 0.62
overall: sigma 1.38, Ppk 0.58
expected below floor: 41,610 ppm
density
yield, %
Early warning on the Dar es Salaam yield series
recorded yield
with a 0.1 point/day decline from day 61
CUSUM alarm
rolling z alarm (none)
Lower CUSUM, in units of sigma
recorded
with decline
alarm at h = 4 sigma
C / sigma

How the notebook is built

It checks the record before it trusts it. The file is read as text, and a triage table lists every cell that is blank, carries a unit or a decimal comma, or has a date in the wrong format, before anything is converted. Two faults do not show up as text: an energy reading logged in MWh and an impossible ambient temperature. The cleaning function fixes or blanks each one, logs it, and leaves the raw table untouched. Of the 91 rows, one is an exact duplicate, and 89 days carry a usable yield.

The exploratory section ends in hypotheses, not just plots. Yield averages 77.39 % with a standard deviation of 1.38 points, and the floor sits less than two standard deviations below the mean. Hall temperature has the strongest link to yield ($r = -0.48$): days above 25 °C averaged 76.61 % against 77.90 % on cooler days. Each hypothesis is picked up again in a later section.

Normality is tested with Shapiro-Wilk ($p = 0.80$ for yield) and Anderson-Darling, with Q-Q plots, and the notebook then puts a number on what non-normality would do to the false-alarm rate. It also shows that the energy series as logged fails the test because of one data error, which is a cleaning job and not a reason to transform.

The control limits are calculated from the constants, not taken from a package. The I-MR limits are fitted on days 1 to 30, giving a centre line of 77.46 % and limits of 73.32 to 81.60 %, and then applied to all 90 days. A second set fitted on every day is compared with them, and the notebook explains why the frozen baseline is the one to keep even though, on this steady run, the two sets barely differ. The one moving-range signal, a jump of 5.36 points into the best day of the run, is treated as a question to ask rather than a fault.

Capability is handled as a one-sided problem. With only a floor, Cp and Pp are undefined, and the notebook says so instead of inventing an upper limit. Cpk is 0.62, from a within sigma of 1.29, and Ppk is 0.58. The normal model expects about 41,600 below-floor days per million; the record shows 6. Each degree of hall temperature above 25 °C costs about 0.52 points of yield, and removing that loss would lift Cpk to about 0.86, which is a large step but not enough on its own.

The early-warning section tunes its thresholds against evidence. Simulated in-control runs give each detector's false-alarm rate, and runs with an injected fault give its detection delay. At the chosen threshold the CUSUM raises 0.24 false alarms per 90 days and catches a slow decline in a median of 14 days, where a 7-day rolling z-score held to the same budget misses most of them. On the real 90 days the CUSUM raises no alarm; on a copy with a decline added from day 61 it alarms 17 days later. A separate entry screen catches bad readings, and a three-level alert table says who acts at each level. The recommendations are written for a quality manager with no statistics.

Why the exemplar uses one dataset

The brief's methods assume a high-frequency sensor series and subgrouped measurements, and the Dar es Salaam record has neither. No committed workshop dataset holds high-frequency data for Dar es Salaam. The only 1-second or minute-level sets in the workshops are the Port Louis spirulina telemetry (W12), which is the brief's own site, and the Cape Town rooibos energy log (W09), which is the A01 exemplar's primary data, and the exemplar specification rules out both. So the notebook applies every method at daily resolution and says what that costs: its X-bar/R chart runs on consecutive 5-day blocks that are not true rational subgroups, and its detectors are tested on faults

injected into real data, since the record contains no real decline. The yield floor is not invented. It is the plant's own definition of a good day in the W06 plant register.

How it maps to the marking scheme

Rubric row	Band it aims at	Why	What would lift it a band
Exploratory Data Analysis	Distinction	Full descriptive statistics for all three measures, a six-panel figure, and two findings a quick look misses: the yield loss starts above a hall temperature of 25 °C, and energy tracks yield on shift A only, which the notebook reports honestly as a lead, not a result. Correlation, lag-1 autocorrelation and stated hypotheses reach into Stretch.	Decompose the series into trend and a weekly cycle, plot the full autocorrelation function, and test each hypothesis formally where it is used, which is what Stretch asks for.
Normality Testing	Distinction	Two tests with Q-Q plots, the false-alarm rate quantified under a fitted skew-normal distribution, percentile limits compared with three-sigma limits, and the failed test on raw energy traced to a data error instead of being transformed away.	Compare Box-Cox and Johnson transformations with before and after Q-Q plots on a series that is genuinely non-normal, and cite food-industry guidance on non-normal process data.
Control Chart Construction	Distinction	I-MR limits worked from the constants with zones and four run rules, written as reusable functions; limits fitted on a phase I window and compared with an all-days fit, with the reason the frozen baseline is kept; the moving-range signal discussed. The X-bar/R chart runs on 5-day blocks, which the notebook admits are weak subgroups.	Interactive plotly charts with hover, the functions moved into an importable monitoring module, and a sensitivity analysis of limit width and block size.

Rubric row	Band it aims at	Why	What would lift it a band
Capability Analysis	Distinction	The one-sided floor is handled correctly: Cp and Pp are stated as undefined with the reason, Cpk and Ppk come with confidence intervals, expected and observed below-floor days are compared, and the target is turned into a required mean or sigma plus a projection of removing the heat loss. A marker who reads "all four indices" strictly may see two missing; the stated reason should carry the band.	A yield-loss analysis in money, which needs throughput and calyx price that the shift return does not hold, and a rolling-capability chart instead of a sentence describing one.
Anomaly Detection	Distinction	Two detectors are compared on one false-alarm budget, with thresholds tuned by simulation for both a slow decline and a sudden step, the chosen CUSUM is shown on real and injected data, a separate entry screen compares a robust z-score with an isolation forest, and deployment cost, latency and a three-level alert table are covered.	Put all three methods through the same detection simulation with one results table, and tie each alert level to a stated consequence and probability, which is the Stretch description.
Code Quality and Narrative	Stretch	A requirements cell, a fixed seed, every analysis choice in one parameter cell, a function for each method, a table of contents, markdown between code cells throughout, and a narrative that reads as a report; the same functions take a 1-second series, so it could serve as a template for other plants.	Within Stretch, move the functions into a tested module the notebook imports and pin the library versions, so a rerun next year gives the same answer.

Where the figures come from

Every figure the notebook states is computed by its own code when `scripts/build_a03_exemplar.py` builds and executes it; the markdown is filled in from those values, and the build stops if the kernel's figures differ from the text. `tests/test_exemplar_a03.py` resolves each locator below against the data without the builder's help, and checks the executed notebook printed the same value. The analysis

choices (the 30-day baseline, the 5-day blocks, the false-alarm budget, the injected decline of a tenth of a point a day) are the author's and have no source.

Figure	Value	Source
Rows in the shift-return file	91	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/dar_es_salaam_hibiscus_90d.csv:rows`
Exact duplicate rows	1	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/dar_es_salaam_hibiscus_90d.csv:duplicates`
Days with a usable yield	89	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/dar_es_salaam_hibiscus_90d.csv:count(yield_pct)`
Yield floor (LSL), %	75	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/generate_dashboard_exemplar.py:UBUNTU_GROUP[tea].good_def`
Mean daily yield, %	77.39	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/dar_es_salaam_hibiscus_90d.csv:mean(yield_pct)`
Standard deviation of yield, points	1.38	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/dar_es_salaam_hibiscus_90d.csv:std(yield_pct)`
Shapiro-Wilk p, yield	0.80	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/dar_es_salaam_hibiscus_90d.csv:shapiro_p(yield_pct)`
Correlation, hall temperature and yield	-0.48	`workshops/06 excel to ai dashboarding/graduate/group assets/dar es salaam hibiscus 90d.csv:corr(ambient_temp_c,yield_pct)`
Mean yield above 25 °C, %	76.61	`workshops/06 excel to ai dashboarding/graduate/group assets/dar es salaam hibiscus 90d.csv:mean_where(yield_pct,ambient_temp_c>25)`
Mean yield at 25 °C or below, %	77.90	`workshops/06 excel to ai dashboarding/graduate/group assets/dar es salaam hibiscus 90d.csv:mean_where(yield_pct,ambient_temp_c<=25)`
I chart centre line, days 1-30, %	77.46	`workshops/06 excel to ai dashboarding/graduate/group assets/dar es salaam hibiscus 90d.csv:imr(yield_pct,days 1-30).cl`

Figure	Value	Source
I chart lower limit, days 1-30, %	73.32	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/dar_es_salaam_hibiscus_90d.csv:imr(yield_pct,days 1-30).lcl`
I chart upper limit, days 1-30, %	81.60	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/dar_es_salaam_hibiscus_90d.csv:imr(yield_pct,days 1-30).ucl`
Largest moving range, points	5.36	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/dar_es_salaam_hibiscus_90d.csv:max_mr(yield_pct)`
Sigma within (MR-bar / 1.128), points	1.29	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/dar_es_salaam_hibiscus_90d.csv:sigma_mr(yield_pct)`
Cpk	0.62	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/dar_es_salaam_hibiscus_90d.csv:cpk(yield_pct,75)`
Ppk	0.58	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/dar_es_salaam_hibiscus_90d.csv:ppk(yield_pct,75)`
Expected below-floor days per million	41,600	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/dar_es_salaam_hibiscus_90d.csv:ppm_below(yield_pct,75)`
Days below the floor	6	`workshops/06 excel to ai dashboarding/graduate/group assets/dar es salaam hibiscus 90d.csv:count_below(yield_pct,75)`
Yield lost per °C above 25 °C, points	0.52	`workshops/06 excel to ai dashboarding/graduate/group assets/dar es salaam hibiscus 90d.csv:heat_loss_per_degree(yield_pct,ambient_temp_c,25)`
Cpk with the heat loss removed	0.86	`workshops/06 excel to ai dashboarding/graduate/group assets/dar es salaam hibiscus 90d.csv:cpk_heat_removed(yield_pct,75,25)`
CUSUM false alarms per 90 days, simulated	0.24	`workshops/06 excel to ai dashboarding/graduate/group assets/dar es salaam hibiscus 90d.csv:cusum_sim(yield_pct,days 1-30,k=0.5,h=4,seed=2026,n=2000,days=90).false_alarms`

Figure	Value	Source
CUSUM median delay on a slow decline, days	14	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/dar_es_salaam_hibiscus_90d.csv:cusum_sim(yield_pct,days 1-30,k=0.5,h=4,seed=2026,n=2000,days=90,decline=0.1/day from day 31).median_delay`
CUSUM alarms on the recorded days	0	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/dar_es_salaam_hibiscus_90d.csv:cusum_alarms(yield_pct,days 1-30,k=0.5,h=4)`
Days from injected decline to CUSUM alarm	17	`workshops/06_excel_to_ai_dashboarding/graduate/group_assets/dar_es_salaam_hibiscus_90d.csv:cusum_delay(yield_pct,days 1-30,k=0.5,h=4,decline=0.1/day from day 61)`